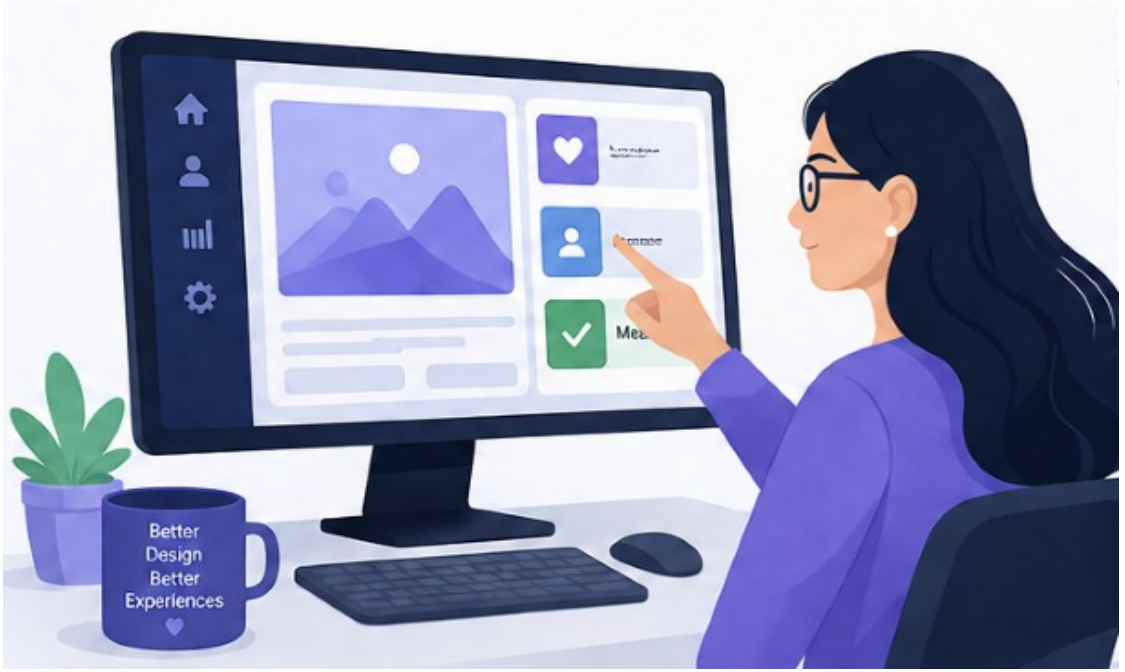
HUMAN COMPUTER INTERACTION

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LEARNING OUTCOMES

After studying this chapter, students should be able to:

Identify different users and stakeholders.

Gather user requirements.

Analyze the context in which a system will be used.

Differentiate between functional and non-functional requirements.

Develop use cases.

Appreciate the importance of user-centered design.

REQUIREMENT ANALYSIS IN HCI

- Requirement analysis is a critical phase in the design and development of any interactive system, particularly in Human-Computer Interaction (HCI). It involves understanding the needs, goals, and constraints of users and stakeholders, which are fundamental to creating an effective, usable, and user-friendly system.

- Unlike traditional software engineering, HCI requirement analysis focuses not only on **what the system should do**, but also **how users interact with it**.

- In HCI, requirement analysis focuses not only on functional requirements (what the system should do) but also on non-

REQUIREMENT ANALYSIS IN HCI

- Proper requirement analysis ensures that the final system aligns with user expectations and provides a positive experience.

KEY COMPONENTS OF REQUIREMENT ANALYSIS IN HCI

1. User Needs Identification

User Analysis: Understanding who the users are, their characteristics, skills, goals, and limitations.

User Tasks: What are the users trying to achieve with the system? These tasks should be defined in terms of user goals, and a distinction should be made between primary (core) and secondary tasks.

User Preferences: Identifying preferences in terms of user interface design, interaction modalities (touch, voice, etc.), and accessibility features.

User Environment: The physical, social, and technological context in which the user will interact with the system, including the devices, locations, and conditions that influence the interaction.

KEY COMPONENTS OF REQUIREMENT ANALYSIS IN HCI

If user needs are misunderstood,

Users become frustrated.

Productivity decreases.

Errors increase.

The system may completely fail.

Example:

Suppose a hospital introduces an electronic medical record syster

The software has 100 features.

Doctors only need 10 of them during emergencies.

If those 10 features are difficult to access,

the software becomes unusable despite having many features.

KEY COMPONENTS OF REQUIREMENT ANALYSIS IN HCI

2. Stakeholder Requirements

Stakeholder Identification: Apart from users, other stakeholders could include business owners, developers, designers, or regulatory bodies. These individuals or groups might have different expectations or requirements.

Stakeholder Goals: Stakeholders may have operational, financial, or legal goals that must be considered (e.g., security, budget constraints, or regulatory compliance).

KEY COMPONENTS OF REQUIREMENT ANALYSIS IN HCI

3. Context of Use: Context of Use describes: Under what conditions will users interact with the system?

Physical Context: Will users be interacting with the system in a stationary office, on the move, or in noisy environments? This helps inform design decisions (e.g., screen readability, input mechanisms).

Social Context: Is the system used by individuals or in collaborative settings? Social dynamics and how users work together should inform interaction design (e.g., multi-user interactions, privacy concerns).

Technological Context: What devices and platforms will the system run on (mobile phones, desktop computers, smartwatches)? This influences the design choices regarding responsiveness, usability, and compatibility.

KEY COMPONENTS OF REQUIREMENT ANALYSIS IN HCI

3. Context of Use

Example: Fitness Tracking App

- **Physical Context**
  - **Stationary (Office/Home):** Users track detailed workout data on a desktop or tablet.
  - **On the Move (Gym/Outdoors):** Users check stats (e.g., heart rate, steps) on their phone during exercise.
  - **Noisy Environments (Outdoor/Busy Gym):** Large text and visual cues for easy readability.

Design Implication: Responsive design for different devices, large fonts for readability, and easy touch interactions for mobile use.

KEY COMPONENTS OF REQUIREMENT ANALYSIS IN HCI

3. Context of Use

Example: Fitness Tracking App

- **Social Context**:
  - **Individual Use:** Users track personal fitness goals and progress.
  - **Collaborative Use:** Users join group challenges, share achievements, and interact with a fitness community.

Design Implication: Privacy controls for individual use, social features for group interactions.

KEY COMPONENTS OF REQUIREMENT ANALYSIS IN HCI

3. Context of Use

Example: Fitness Tracking App

- **Technological Context**:
  - **Mobile Phones:** Quick access to real-time workout tracking.
  - **Desktop:** Detailed analytics and long-term progress reports.
  - **Smartwatches:** Glanceable stats during workouts (e.g., heart rate, calories).

Design Implication: Different interfaces for mobile, desktop, and smartwatch, optimizing for ease of use across devices.

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KEY COMPONENTS OF REQUIREMENT ANALYSIS IN HCI

4. Functional Requirements

This defines the system's core functionalities—what the system must do to meet user goals.

5. Non-functional Requirements

These specify the system's quality attributes, including:

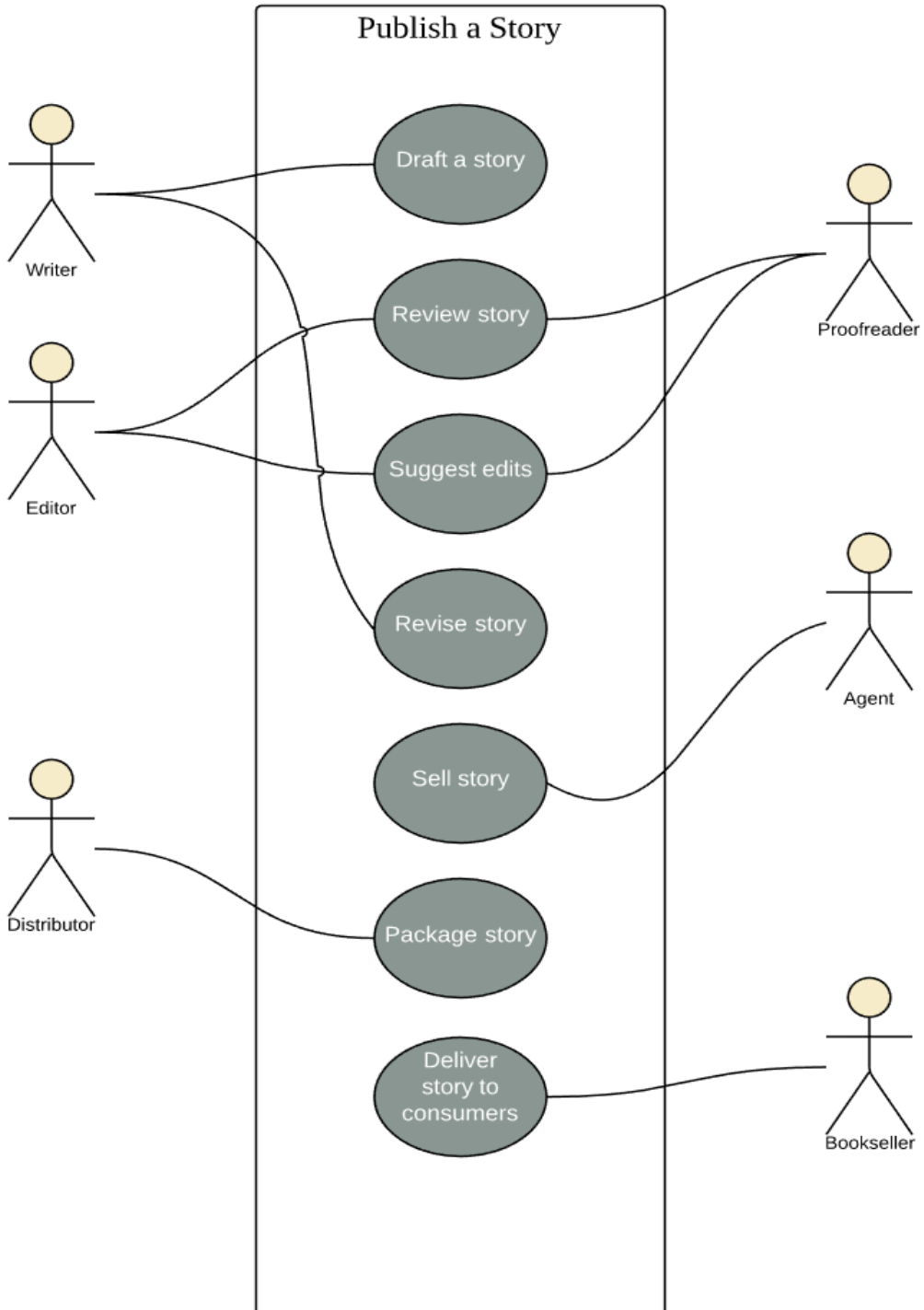
- **Performance**: How fast or responsive the system should be.
- **Usability**: How easy it should be for users to learn and use the system.
- **Reliability**: The system's ability to perform consistently under different conditions.
- **Security and Privacy**: Protection of user data and information.
- **Accessibility:** Ensuring that users with disabilities can use the system effectively.

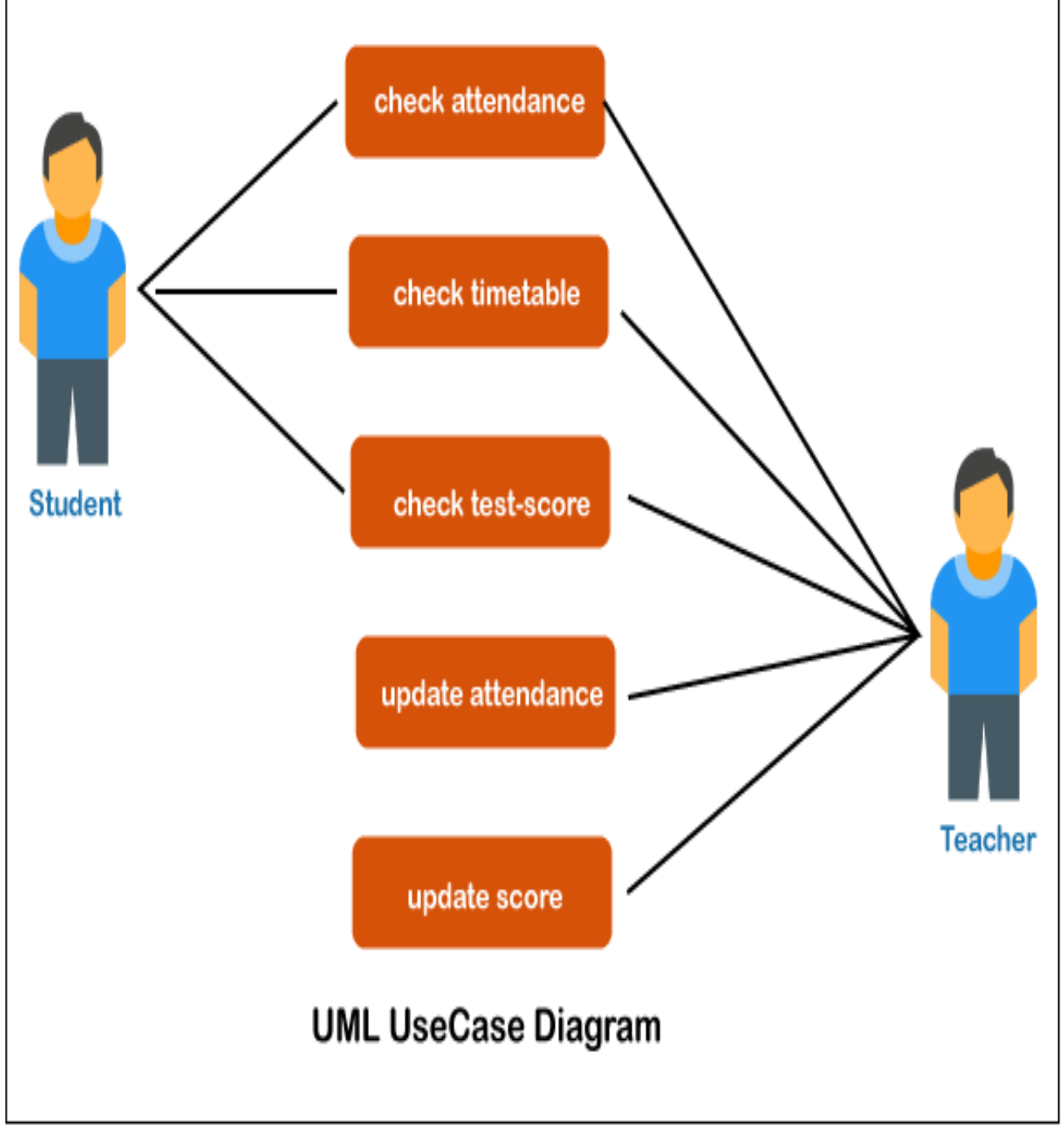
KEY COMPONENTS OF REQUIREMENT ANALYSIS IN HCI

6. Use Case Development

- Use cases are scenarios that describe how users will interact with the system to accomplish specific tasks.
- These help in understanding the system's functional requirements and user needs in more detail

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Publish a Story
Writer
Editor
Distributor
Draft a story
Review story
Suggest edits
Revise story
Sell story
Package story
Deliver story to consumers
Proofreader
Agent
Bookseller


check attendance
check timetable
check test-score
update attendance
update score
Student
Teacher
UML UseCase Diagram

CASE STUDY: SMART UNIVERSITY CAFETERIA ORDERING SYSTEM

Scenario

The university cafeteria experiences long queues during lunch hours. Students often wait 20–30 minutes, some menu items sell out before they reach the counter, and staff struggle to manage peak demand. The university wants a mobile application that allows users to browse menus, place orders, pay digitally, and receive notifications when their food is ready.

CASE STUDY: SMART UNIVERSITY CAFETERIA ORDERING SYSTEM

Step 1: Identify Users

Students

Faculty

Cafeteria staff

Cashier

Cafeteria manager

Delivery staff (optional)

Step 2: Identify User Needs

Students:

Quick ordering

Short waiting time

Digital payment

Order tracking

Favorite meals

Staff:

Easy order management

Clear preparation queue

Step 2: Identify User Needs

Manager:

Sales reports

Inventory tracking

Peak-hour analytics

3: Stakeholder Requirements

Stakeholder	Requirement
Student	Order food quickly
Faculty	Priority pickup option
Staff	View and update order status
Manager	Generate sales and inventory reports
University	Secure payment records

Step 4: Context of Use

- **Users:** Students, faculty, cafeteria staff
- **Tasks:** Browse menu, order food, pay, collect order
- **Environment:** Busy cafeteria, noisy surroundings, variable internet connectivity
- **Devices:** Mostly smartphones, some tablets at counters

Step 5: Functional Requirements

- User registration and login
- Browse menu
- Search food items
- Place order
- Online payment
- Order tracking
- Push notifications
- Feedback and ratings

Step 6: Non-Functional Requirements

Order confirmation within 2 seconds
99.9% availability during semester
Simple interface with minimal steps
High-contrast mode and readable fonts
Secure payment and encrypted user data

Step 7: Use Case Example

- In-Class Activity

METHODS AND TECHNIQUES FOR REQUIREMENT ANALYSIS IN HCI

1. **Interviews and Surveys:** Talk to users and stakeholders to understand their needs and behaviors. **Example:** Interview users of a fitness app to understand what features they like and what's missing. Use a survey to gather feedback from a larger group on app usability.
2. **Observations:** Watch users in their natural environment to find out what they really need. **Example:** Observe users at a gym using the fitness app. Notice if they struggle to track their workout or if they often get distracted by complicated features.
3. **Focus Groups:** Group discussions to identify common problems and ideas for improvement. **Example:** Host a group of fitness app users to discuss their experience. Ask about the features they find most helpful and what changes would improve their experience.
4. **Task Analysis:** Break down the steps users take to complete tasks and find obstacles. **Example:** Analyze the steps a user takes to log a workout in the app, from opening the app to entering their data, identifying where they might get confused or frustrated.

METHODS AND TECHNIQUES FOR REQUIREMENT ANALYSIS IN HCI

5. Personas: Create fictional characters that represent different user types to guide design.

6. Scenarios and Prototyping: Use stories and prototypes to test and improve design ideas.

Example: Create a persona for "Sarah, a busy professional," who needs to log her workouts quickly. In her scenario, she's at work and wants to quickly log a 10-minute workout during her break. Based on this, design a simple prototype with a fast, one-tap feature for logging workouts, and test it to ensure it fits her need for convenience and speed.

STEPS IN REQUIREMENT ANALYSIS

- **Define the Problem:** Understand the problem or opportunity, gather insights from stakeholders, and identify goals and constraints.

- **Identify Stakeholders and Users:** Identify stakeholders (e.g., users, clients) and define their roles, needs, and expectations.

- **Elicit Requirements:** Gather user and stakeholder requirements through interviews, surveys, and task analysis.

- **Prioritize Requirements:** Prioritize requirements based on importance and feasibility.

- **Analyze and Document Requirements:** Identify gaps or conflicts and create a document outlining functional and user requirements.

- **Validate Requirements:** Confirm with stakeholders and users to ensure requirements meet real needs, through reviews or usability testing.

COMMON CHALLENGES IN REQUIREMENT ANALYSIS

- **Miscommunication:** Users may struggle to articulate needs, leading to conflicting requirements.
- **Changing Requirements:** New needs may emerge, complicating project scope management.
- **Diverse User Needs:** Different users (e.g., novices, experts, disabled users) can have conflicting requirements.